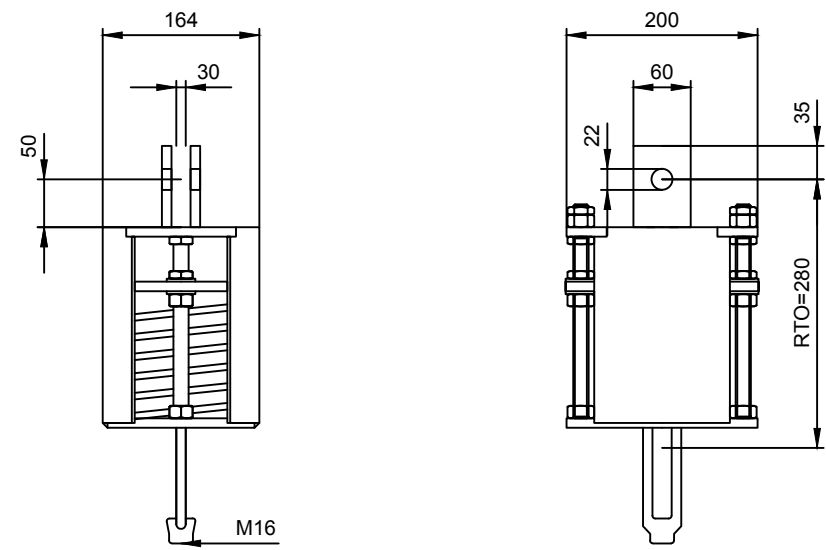
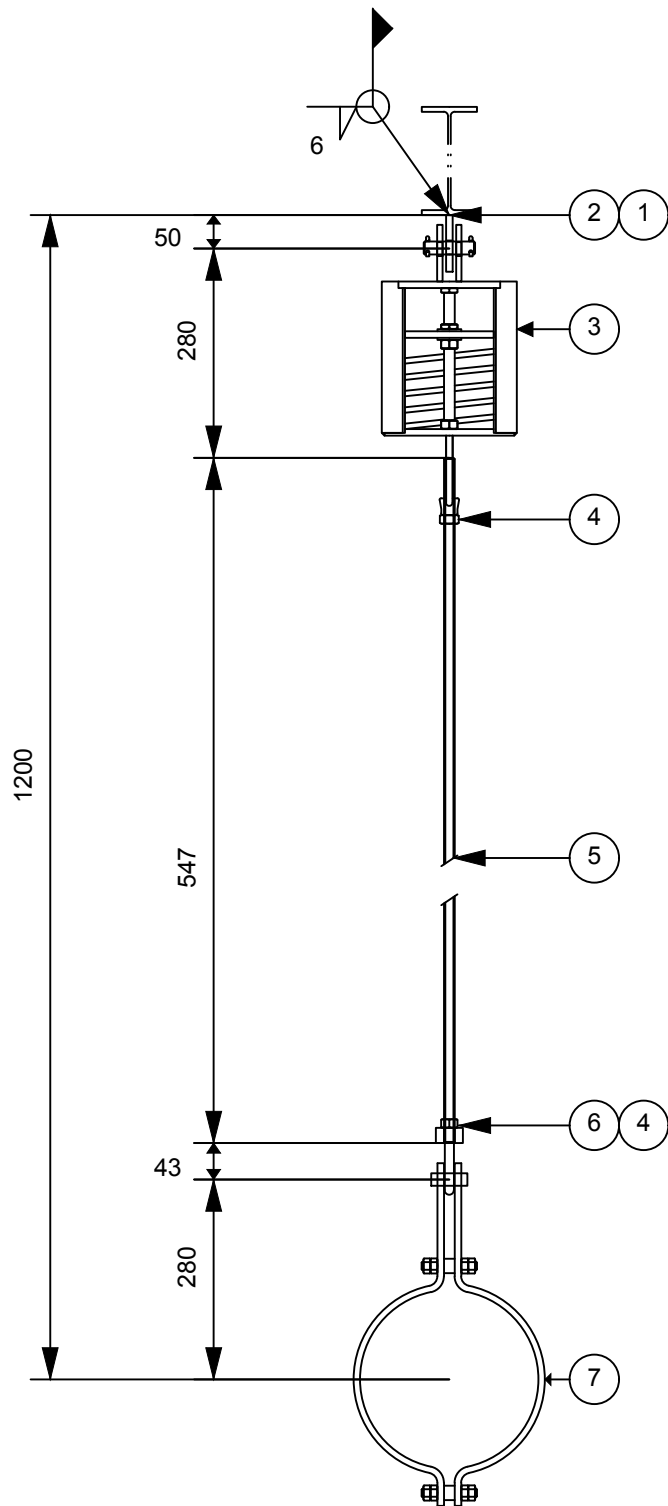
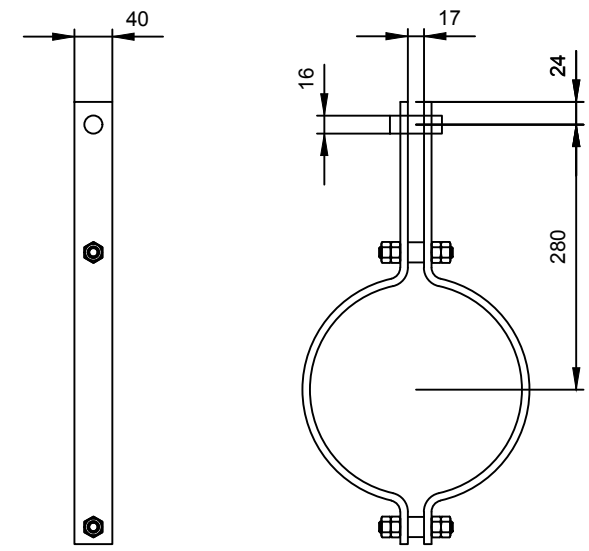


Mass of Support Assembly:17.59kg



DETAIL ITEM NO: 3



DETAIL ITEM NO: 7

Item No	No Reqd	Figure No	Description	Material
1	1	UA1-M20	Lug Attachment	IS:2062 E250.Gr.BR
2	1	RA14-M20	Clevis Pin	IS:2062 E250.Gr.BR
3	1	VS1-16-C	Variable Effort Support	Per MSS-SP58
4	2	RA4-M16	Lock Nut	IS: 1367 Cl.4.0
5	1	RA3-M16 L=547	All Thread Rod RH/RH	IS:2062 E250.Gr.BR
6	1	RA15-M16	Weldless Eye Nut	ASTM A105
7	1	PA3-200-1-400	Pipe Clamp 3 Bolt	IS:2062 E250.Gr.BR

Stud bolts and nuts	A193 GR.B7 /A194 GR.2H
Spacer tube	A 106 Gr.B

SpringDetails
VS1-16 C Preset LoadInc Anc wt.=4295N(438.0kg).Operating LoadInc Anc wt.=4039N(411.9kg)
Spring Rate=72.9N/mm. RTO=280mm. Pin=M20,Rod = M16.Thermal Movement 3.52mm. Up Variability=6.34%
Rod Angulation = 0.14 Deg

- NOTES:
1. ALL DIMENSIONS ARE IN MM ONLY UNLESS OTHERWISE STATED.
 2. ASSEMBLY LENGTH ONLY WILL BE OFFERED FOR INSPECTION.
 3. ASSEMBLY LENGTH CAN BE ADJUSTED ± 75 MM BY TURNBUCKLE.

PAINTING:
1. KINDLY REFER OUR DOCUMENT NO.8772-FS-001.

Pipe NB : 200	Temperature : 35°C	Pipe Insulation : -
Loads : -	Movement : -	Vertical mm Horizontal mm
Pipe Load : 4000.0N	Preset to operating :	3.515 Up 2.113

REVISIONS						
	1	03-12-25	CLIENT COMMENTS INCORPORATED	CVN	JSR	JSR
	0	21-11-25	ISSUED FOR APPROVAL	CVN	JSR	JSR
REV	DATE	DESCRIPTION			PREP	CHD APPD

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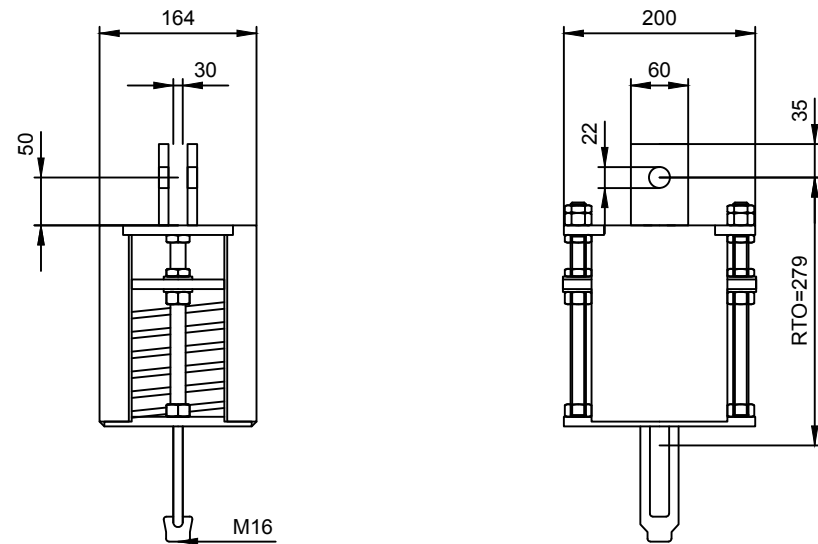
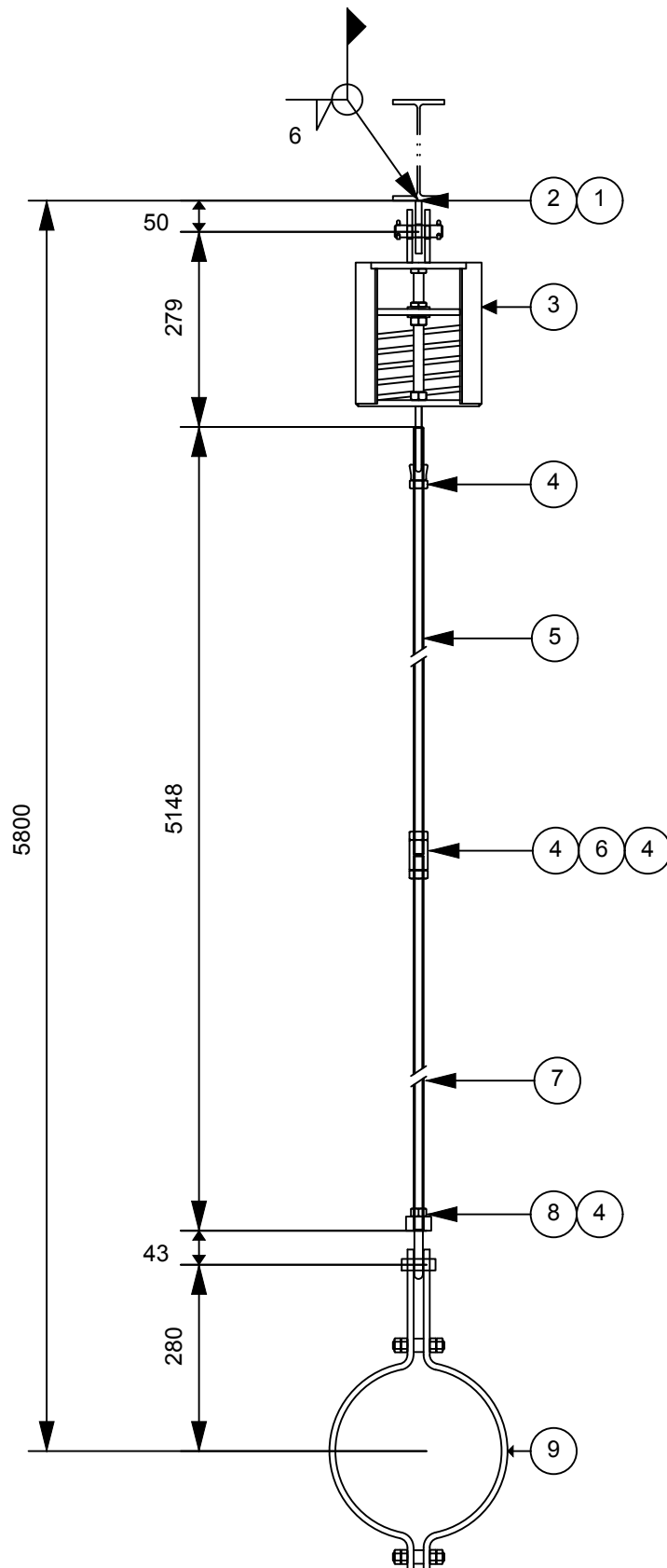
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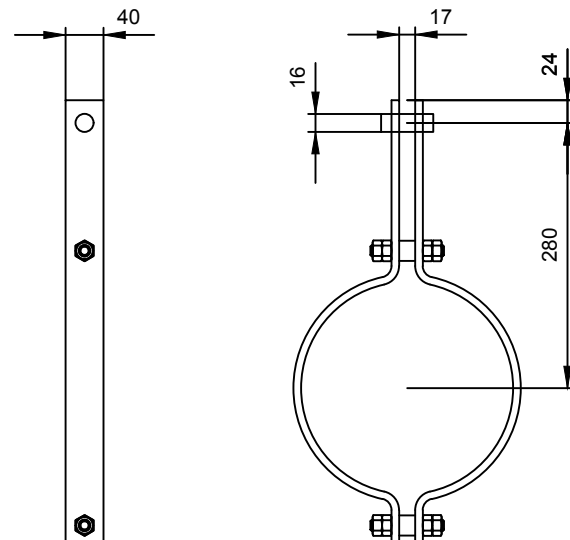
PIPE HANGERS&SUPPORTS
PVT LTD.
29 INDUSTRIAL ESTATE PERUNGUDI
CHENNAI 600096
INDIA

SYSTEM NO\ NODE NO: EM374-CN-010A/2075		CUSTOMER YARA FERTILIZERS INDIA PVT LTD	
LINE NO. 8"-CW-1A-402		PROJECT -	
TITLE: SUPPORT ASSEMBLY		PIPE SYSTEM -	
SUPPORT MARK No: SP-009		CUSTOMER ORDER NO. 4110000507 DT: 17.11.2025	
No ASSEMBLIES 1		CONTRACT No. 8772	
DRAWING No 8772-03		FINISH SEE FINISH PROCEDURE	
DATE 21-11-25		THREAD FORM ISO-METRIC COARSE	
REV 1		SCALE NTS	
DATE 21-11-25		DRAWN CVN	
DATE 21-11-25		CHECKED JSR	

Mass of Support Assembly:23.53kg



DETAIL ITEM NO: 3



DETAIL ITEM NO: 9

Item No	No Reqd	Figure No	Description	Material
1	1	UA1-M20	Lug Attachment	IS:2062 E250.Gr.BR
2	1	RA14-M20	Clevis Pin	IS:2062 E250.Gr.BR
3	1	VS1-16-C	Variable Effort Support	Per MSS-SP58
4	4	RA5-M16	Lock Nut	IS: 1367 Cl.4.0
5	1	RA3-M16 L=3000	All Thread Rod RH/RH	IS:2062 E250.Gr.BR
6	1	RA7-M16	Rod Coupling	IS:2062 E250.Gr.BR
7	1	RA3-M16 L=2148	All Thread Rod RH/RH	IS:2062 E250.Gr.BR
8	1	RA15-M16	Weldless Eye Nut	ASTM A105
9	1	PA3-200-1-400	Pipe Clamp 3 Bolt	IS:2062 E250.Gr.BR

Stud bolts and nuts	A193 GR.B7 /A194 GR.2H
Spacer tube	A 106 Gr.B

SpringDetails

VS1-16 C Preset LoadInc Anc wt.=4256N(434.0kg).Operating LoadInc Anc wt.=4097N(417.8kg)
Spring Rate=72.9N/mm. RTO=279mm. Pin=M20,Rod = M16.Thermal Movement 2.18mm. Up Variability=3.88%
Rod Angulation = 0.03 Deg

NOTES:

1. ALL DIMENSIONS ARE IN MM ONLY UNLESS OTHERWISE STATED.
2. ASSEMBLY LENGTH ONLY WILL BE OFFERED FOR INSPECTION.
3. ASSEMBLY LENGTH CAN BE ADJUSTED ± 75 MM BY TURNBUCKLE.

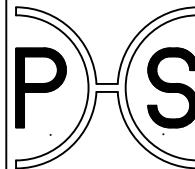
PAINTING:

1. KINDLY REFER OUR DOCUMENT NO.8772-FS-001.

Pipe NB : 200	Temperature : 44°C	Pipe Insulation : -	
Loads : -	Movement : -	Vertical mm	Horizontal mm
Pipe Load : 4000.0N	Preset to operating :	2.18 Up	3.135

REVISIONS						
	1	03-12-25	CLIENT COMMENTS INCORPORATED	CVN	JSR	JSR
	0	21-11-25	ISSUED FOR APPROVAL	CVN	JSR	JSR
	REV	DATE	DESCRIPTION	PREP	CHD	APPD

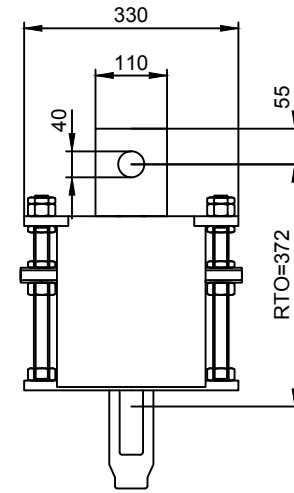
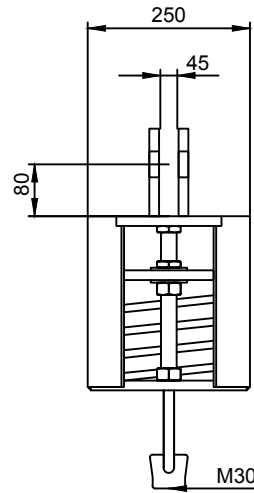
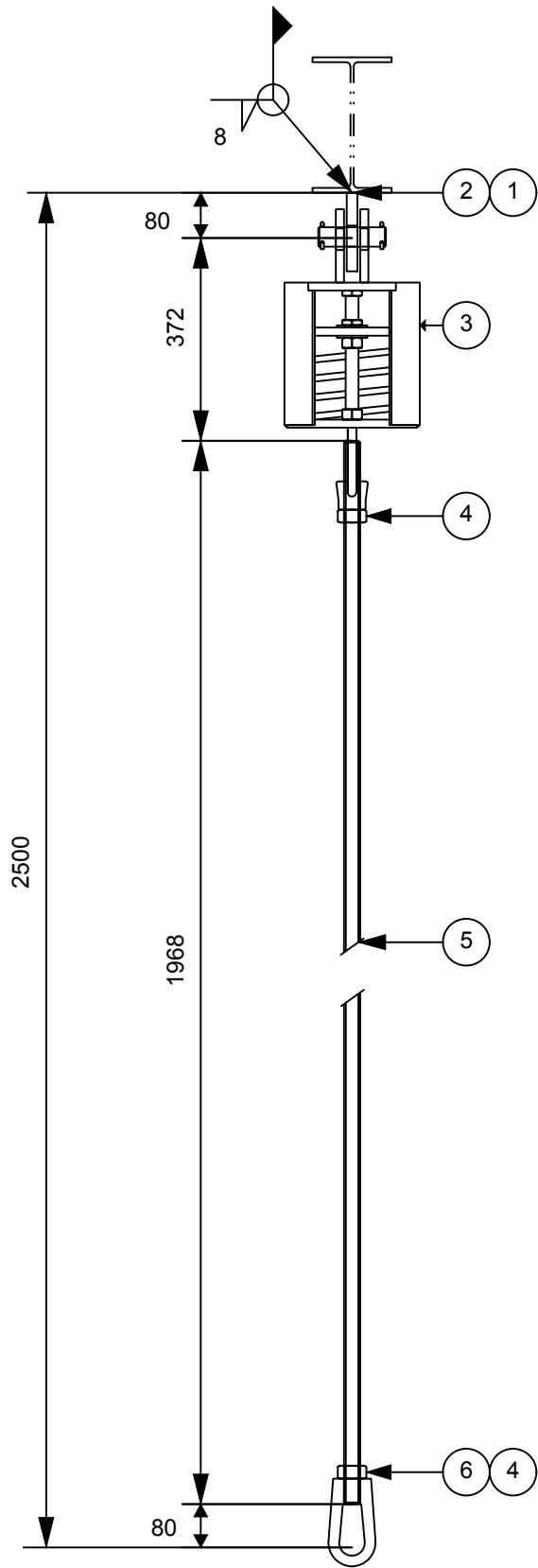
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CHENNAI 600096
INDIA

SYSTEM NO\ NODE NO:	EM374-CN-010A/3075	FINISH	SEE FINISH PROCEDURE		
LINE NO.	8"-CW-1A-403	THREAD FORM	ISO-METRIC COARSE		
		SCALE	NTS		
TITLE:		SUPPORT ASSEMBLY	DRAWN	CVN	DATE 21-11-25
SUPPORT MARK No:		SP-010	CHECKED	JSR	DATE 21-11-25
No ASSEMBLIES 1		DRAWING No 8772-04			REV 1

Mass of Support Assembly:62.62kg



DETAIL ITEM NO: 3

Item No	No Reqd	Figure No	Description	Material
1	1	UA1-M36	Lug Attachment	IS:2062 E250.Gr.BR
2	1	RA14-M36	Clevis Pin	IS:2062 E250.Gr.BR
3	1	VS1-22-C	Variable Effort Support	Per MSS-SP58
4	2	RA4-M30	Lock Nut	IS: 1367 CI.4.0
5	1	RA3-M30 L=1968	All Thread Rod RH/RH	IS:2062 E250.Gr.BR
6	1	RA15-M30	Weldless Eye Nut	ASTM A105

SpringDetails
VS1-22 C Preset LoadInc Anc wt.=25550N(2605.4kg).Operating LoadInc Anc wt.=23241N(2369.9kg)
Spring Rate=384N/mm. RTO=372mm. Pin=M36,Rod = M30.Thermal Movement 6.01mm. Up Variability=9.93%
Rod Angulation = 0.24 Deg

NOTES:

1. ALL DIMENSIONS ARE IN MM ONLY UNLESS OTHERWISE STATED.
2. ASSEMBLY LENGTH ONLY WILL BE OFFERED FOR INSPECTION.
3. ASSEMBLY LENGTH CAN BE ADJUSTED ± 75 MM BY TURNBUCKLE.

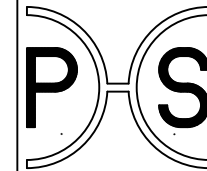
PAINTING:

1. KINDLY REFER OUR DOCUMENT NO.8772-FS-001.

Pipe NB : -	Temperature : 200°C	Pipe Insulation : -	
Loads : -	Movement : -	Vertical mm	Horizontal mm
Pipe Load : 23132.0N	Preset to operating :	6.015 Up	10.201

REVISIONS						
	REV	DATE	DESCRIPTION	PREP	CHD	APPD
	0	24-11-25	ISSUED FOR APPROVAL	CVN	JSR	JSR

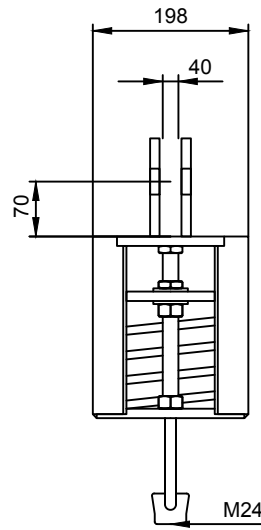
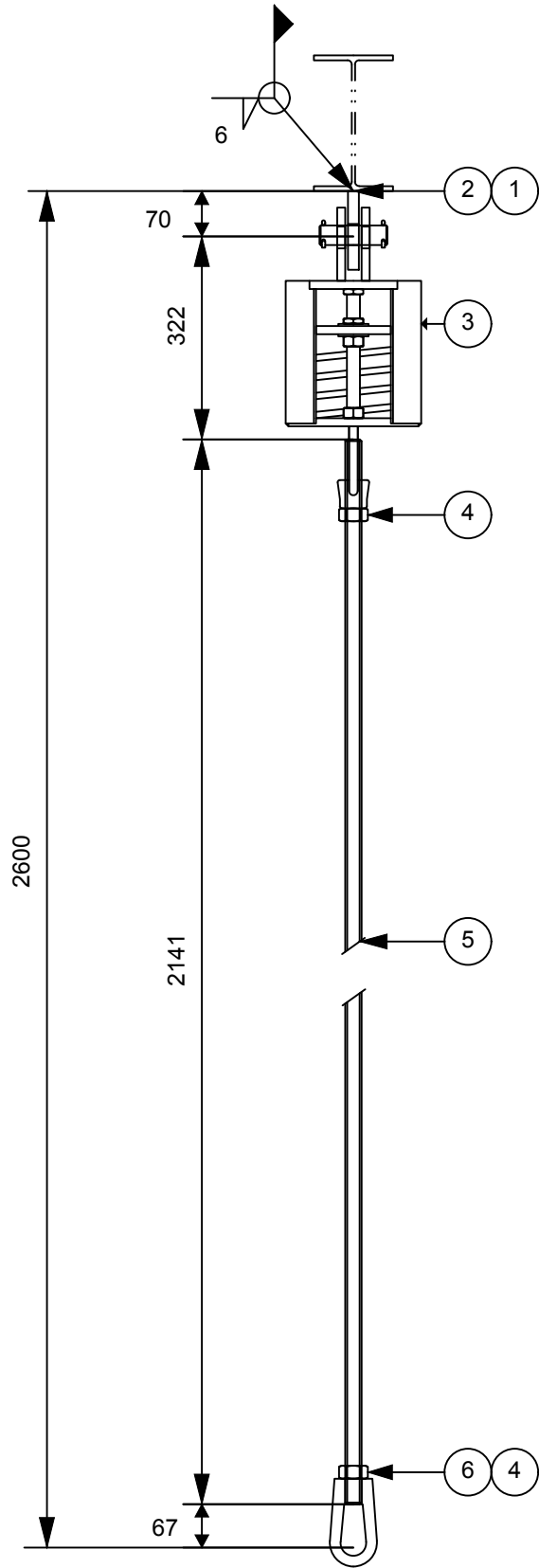
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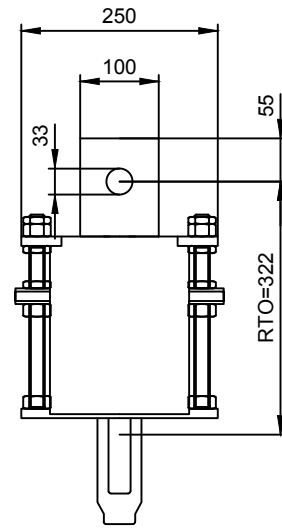
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29 INDUSTRIAL ESTATE PERUNGUDI
CHENNAI 600096
INDIA

SYSTEM NO\ NODE NO:	EM374-CN-010B/4099	FINISH	SEE FINISH PROCEDURE	
LINE NO.	36"-AR-1A-201	THREAD FORM	ISO-METRIC COARSE	
		SCALE	NTS	
TITLE:	SUPPORT ASSEMBLY	DRAWN	CVN	DATE 24-11-25
SUPPORT MARK No:	SP-011	CHECKED	JSR	DATE 24-11-25
No ASSEMBLIES	1	DRAWING No	8772-05	REV 0

Mass of Support Assembly:34.11kg



DETAIL ITEM NO: 3



Item No	No Reqd	Figure No	Description	Material
1	1	UA1-M30	Lug Attachment	IS:2062 E250.Gr.BR
2	1	RA14-M30	Clevis Pin	IS:2062 E250.Gr.BR
3	1	VS1-20-C	Variable Effort Support	Per MSS-SP58
4	2	RA4-M24	Lock Nut	IS: 1367 CI.4.0
5	1	RA3-M24 L=2141	All Thread Rod RH/RH	IS:2062 E250.Gr.BR
6	1	RA15-M24	Weldless Eye Nut	ASTM A105

SpringDetails
VS1-20 C Preset LoadInc Anc wt.=13483N(1374.9kg).Operating LoadInc Anc wt.=14264N(1454.5kg)
Spring Rate=213N/mm. RTO=322mm. Pin=M30,Rod = M24.Thermal Movement 3.67mm. Down Variability=5.47%
Rod Angulation = 0.07 Deg

NOTES:

1. ALL DIMENSIONS ARE IN MM ONLY UNLESS OTHERWISE STATED.
2. ASSEMBLY LENGTH ONLY WILL BE OFFERED FOR INSPECTION.
3. ASSEMBLY LENGTH CAN BE ADJUSTED ± 75 MM BY TURNBUCKLE.

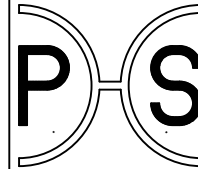
PAINTING:

1. KINDLY REFER OUR DOCUMENT NO.8772-FS-001.

Pipe NB : -	Temperature : 200°C	Pipe Insulation : -	
Loads : -	Movement : -	Vertical mm	Horizontal mm
Pipe Load : 14196.0N	Preset to operating :	3.667 Down	3.114

REVISIONS						
	REV	DATE	DESCRIPTION	PREP	CHD	APPD
	0	24-11-25	ISSUED FOR APPROVAL	CVN	JSR	JSR

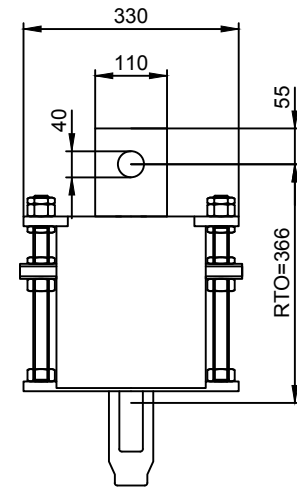
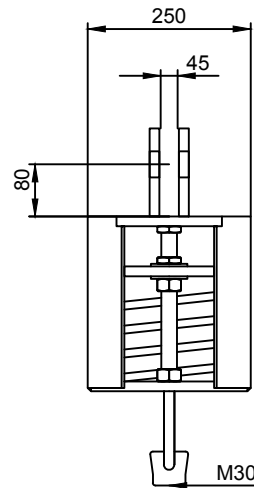
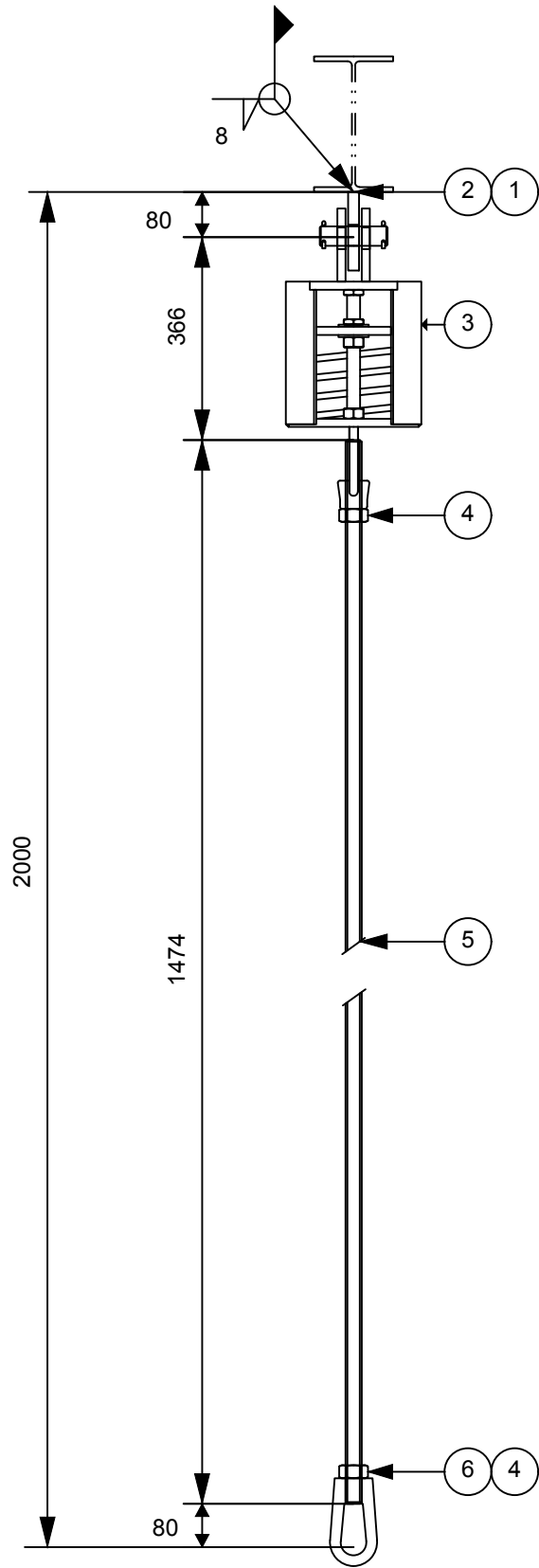
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INDIA

SYSTEM NO\ NODE NO:	EM374-CN-010B/4147	FINISH	SEE FINISH PROCEDURE	
LINE NO.	36"-AR-1A-201	THREAD FORM	ISO-METRIC COARSE	
		SCALE	NTS	
TITLE:	SUPPORT ASSEMBLY	DRAWN	CVN	DATE 24-11-25
SUPPORT MARK No:	SP-012	CHECKED	JSR	DATE 24-11-25
No ASSEMBLIES	1	DRAWING No	8772-06	REV 0

Mass of Support Assembly:60.41kg



DETAIL ITEM NO: 3

Item No	No Reqd	Figure No	Description	Material
1	1	UA1-M36	Lug Attachment	IS:2062 E250.Gr.BR
2	1	RA14-M36	Clevis Pin	IS:2062 E250.Gr.BR
3	1	VS1-22-C	Variable Effort Support	Per MSS-SP58
4	2	RA4-M30	Lock Nut	IS: 1367 Cl.4.0
5	1	RA3-M30 L=1474	All Thread Rod RH/RH	IS:2062 E250.Gr.BR
6	1	RA15-M30	Weldless Eye Nut	ASTM A105

SpringDetails
VS1-22 C Preset LoadInc Anc wt.=23018N(2347.2kg).Operating LoadInc Anc wt.=22187N(2262.5kg)
Spring Rate=384N/mm. RTO=366mm. Pin=M36,Rod = M30.Thermal Movement 2.16mm. Up Variability=3.74%
Rod Angulation = 0.03 Deg

NOTES:

1. ALL DIMENSIONS ARE IN MM ONLY UNLESS OTHERWISE STATED.
2. ASSEMBLY LENGTH ONLY WILL BE OFFERED FOR INSPECTION.
3. ASSEMBLY LENGTH CAN BE ADJUSTED ± 75 MM BY TURNBUCKLE.

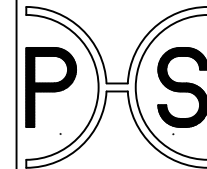
PAINTING:

1. KINDLY REFER OUR DOCUMENT NO.8772-FS-001.

Pipe NB : -	Temperature : 45°C	Pipe Insulation : -	
Loads : -	Movement : -	Vertical mm	Horizontal mm
Pipe Load : 22100.0N	Preset to operating :	2.163 Up	1.034

REVISIONS						
	REV	DATE	DESCRIPTION	PREP	CHD	APPD
	0	24-11-25	ISSUED FOR APPROVAL	CVN	JSR	JSR

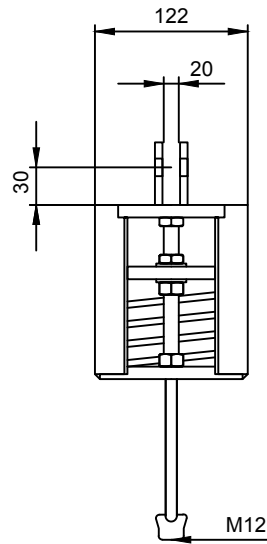
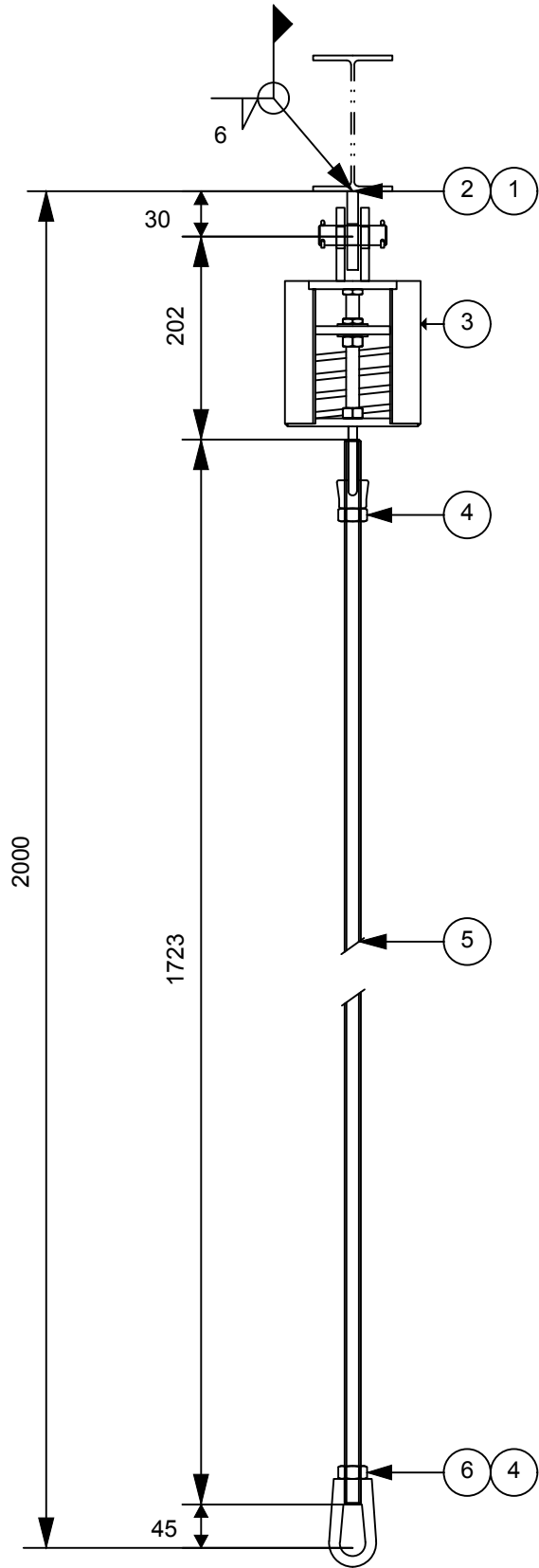
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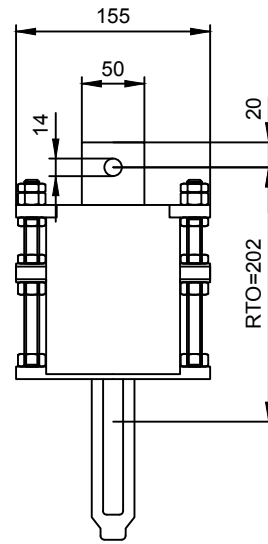
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29 INDUSTRIAL ESTATE PERUNGUDI
CHENNAI 600096
INDIA

SYSTEM NO\ NODE NO:	EM374-CN-010C/5080	FINISH	SEE FINISH PROCEDURE	DATE	24-11-25
LINE NO.	30"-AR-1A-202	THREAD FORM	ISO-METRIC COARSE	DATE	24-11-25
TITLE:	SUPPORT ASSEMBLY	DRAWN	CVN	DATE	24-11-25
SUPPORT MARK No:	SP-013	CHECKED	JSR	DATE	24-11-25
No ASSEMBLIES	1	DRAWING No	8772-07	REV	0

Mass of Support Assembly:6.49kg



DETAIL ITEM NO: 3



Item No	No Reqd	Figure No	Description	Material
1	1	UA1-M12	Lug Attachment	IS:2062 E250.Gr.BR
2	1	RA14-M12	Clevis Pin	IS:2062 E250.Gr.BR
3	1	VS1-11-C	Variable Effort Support	Per MSS-SP58
4	2	RA4-M12	Lock Nut	IS: 1367 Cl.4.0
5	1	RA3-M12 L=1723	All Thread Rod RH/RH	IS:2062 E250.Gr.BR
6	1	RA15-M12	Weldless Eye Nut	ASTM A105

SpringDetails
VS1-11 C Preset LoadInc Anc wt.=1037N(105.7kg).Operating LoadInc Anc wt.=1014N(103.4kg)
Spring Rate=17.0N/mm. RTO=202mm. Pin=M12,Rod = M12.Thermal Movement 1.32mm. Up Variability=2.20%
Rod Angulation = 0.03 Deg

NOTES:

1. ALL DIMENSIONS ARE IN MM ONLY UNLESS OTHERWISE STATED.
2. ASSEMBLY LENGTH ONLY WILL BE OFFERED FOR INSPECTION.
3. ASSEMBLY LENGTH CAN BE ADJUSTED ± 75 MM BY TURNBUCKLE.

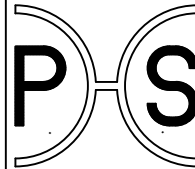
PAINTING:

1. KINDLY REFER OUR DOCUMENT NO.8772-FS-001.

Pipe NB : -	Temperature : 45°C	Pipe Insulation : -	
Loads : -	Movement : -	Vertical mm	Horizontal mm
Pipe Load : 1000.0N	Preset to operating :	1.316 Up	1.046

REVISIONS						
	0	24-11-25	ISSUED FOR APPROVAL		CVN	JSR
	REV	DATE	DESCRIPTION		PREP	CHD

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CHENNAI 600096
INDIA

SYSTEM NO\ NODE NO:	EM374-CN-010C/5090	FINISH	SEE FINISH PROCEDURE
LINE NO.	30"-AR-1A-202	THREAD FORM	ISO-METRIC COARSE
TITLE:	SUPPORT ASSEMBLY	SCALE	NTS
SUPPORT MARK No:	SP-014	DRAWN	CVN
No ASSEMBLIES	1	CHECKED	JSR
DRAWING No	8772-08	DATE	24-11-25
		DATE	24-11-25
		REV	0